PROJECT REPORT
Electricity Consumption Analysis

1. INTRODUCTION

1.1 Project Overview

Electricity Consumption Analysis is a data analytics project designed to analyze electricity usage across different states and regions of India. The project utilizes MySQL for data storage and SQL analysis, Tableau Desktop for creating interactive dashboards, Tableau Public for publishing dashboards, and Flask for deploying the dashboard as a web application.
The project enables users to explore electricity consumption through state-wise analysis, region-wise comparisons, monthly trends, geographical maps, and top electricity-consuming states. Interactive dashboards and story visualizations provide meaningful insights that support data-driven decision-making.

1.2 Purpose

The purpose of this project is to provide an efficient and interactive platform for analyzing electricity consumption data. The project helps users understand consumption patterns, compare electricity usage across different regions, identify high-demand areas, and support better planning and resource allocation through visualization and business intelligence tools.

2. IDEATION PHASE

2.1 Problem Statement

Electricity consumption data generated across different states and regions is large and complex. Manual analysis of such data is time-consuming and often fails to provide meaningful insights.
This project addresses these challenges by developing an interactive Electricity Consumption Analysis system using MySQL, Tableau, and Flask. The solution enables efficient data storage, analysis, visualization, and web-based dashboard access to support informed decision-making.

2.2 Empathy Map Canvas

Think and Feel

- Need accurate electricity consumption analysis.
- Want interactive dashboards.
- Need quick access to electricity usage information.

Hear

- Government energy reports.
- Recommendations from analysts.
- Electricity department feedback.

See

- Different electricity usage across states.
- Seasonal consumption trends.

- Regional variations.

Say and Do

- Analyze electricity consumption.
- Compare states and regions.
- Monitor monthly trends.

Pain

- Manual analysis is slow.
- Large datasets are difficult to understand.
- Traditional reports lack visualization.

Gain

- Faster analysis.
- Interactive dashboards.
- Better planning.
- Improved decision-making.

2.3 Brainstorming

Several project ideas were discussed, including:

- Electricity Consumption Analysis
- Smart Grid Monitoring
- Renewable Energy Analysis
- Energy Demand Forecasting
- Regional Electricity Monitoring

After evaluation, **Electricity Consumption Analysis** was selected because it offers meaningful business insights and demonstrates practical applications of SQL, Tableau, and Flask.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. Identify electricity consumption analysis requirements.
2. Collect electricity consumption data.
3. Store data in MySQL.
4. Perform SQL analysis.
5. Create Tableau visualizations.
6. Publish the dashboard.
7. Access insights through the Flask application.

3.2 Solution Requirements

Functional Requirements

- Import electricity consumption dataset.
- Store data in MySQL.
- Execute SQL queries.
- Create Tableau visualizations.
- Develop interactive dashboards.
- Create story pages.
- Publish the dashboard.

- Integrate with Flask.

Non-Functional Requirements

- Usability
- Reliability
- Performance
- Availability
- Scalability
- Maintainability

3.3 Data Flow Diagram

Electricity Dataset (CSV)
|
▼
MySQL Database
|
▼
SQL Queries
|
▼
Tableau Desktop
|
▼
Interactive Dashboard
|
▼
Tableau Public
|
▼
Flask Web Application
|
▼
End User

3.4 Technology Stack

Component	**Technology Used**
Dataset	CSV File
Database	MySQL
SQL Tool	MySQL Workbench
Data Visualization	Tableau Desktop
Dashboard Publishing	Tableau Public
Programming Language	Python
Web Framework	Flask
IDE	Visual Studio Code
Version Control	GitHub
Documentation	Microsoft Word & PDF

4. PROJECT DESIGN

4.1 Problem Solution Fit

Customer Segment

- Government organizations
- Electricity Boards
- Utility Companies
- Energy Analysts
- Researchers

Customer Constraints

- Large datasets
- Manual analysis
- Lack of visualization

Available Solutions

- Excel
- Manual reports
- SQL reports

Proposed Solution

Interactive Tableau dashboards integrated with MySQL and Flask.

4.2 Proposed Solution

The project stores electricity consumption data in MySQL, performs SQL-based analysis, creates interactive Tableau dashboards, publishes them on Tableau Public, and embeds them into a Flask web application for web access.

4.3 Solution Architecture

Electricity Consumption Dataset (CSV)
|
▼
MySQL Database
|
▼
SQL Query Analysis
|
▼
Tableau Desktop
|
▼
Interactive Charts & Maps
|
▼
Interactive Dashboard
|
▼
Story Creation
|

▼
Tableau Public
|
▼
Flask Web Application
|
▼
End User

5. PROJECT PLANNING & SCHEDULING

Week	Activity
Week 1	Project topic selection and problem identification
Week 2	Dataset collection and preprocessing
Week 3	MySQL database creation and SQL query execution
Week 4	Tableau visualization development
Week 5	Dashboard and Story creation
Week 6	Flask web application development and Tableau Public publishing
Week 7	Testing, GitHub repository creation, and project validation
Week 8	Final documentation, report preparation, and project submission

6. FUNCTIONAL AND PERFORMANCE TESTING

Accuracy Testing

Verified SQL query results and dashboard values against the source dataset.

Performance Testing

Evaluated dashboard loading speed, SQL query execution, and Flask application responsiveness.

Usability Testing

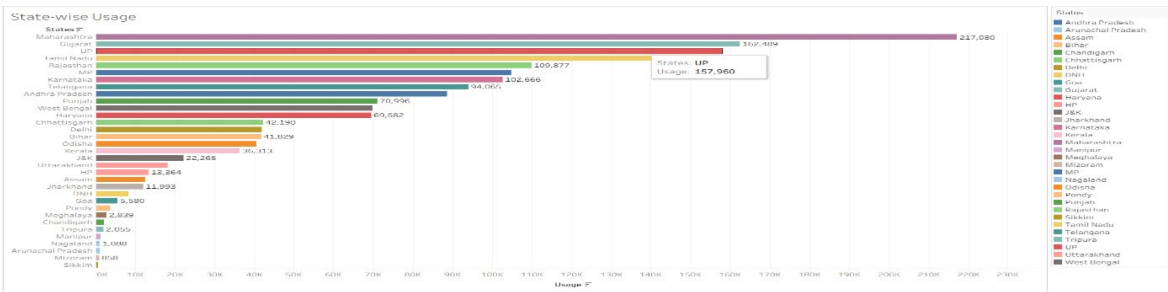
Ensured easy navigation, interactive visualizations, and user-friendly dashboard design.

Compatibility Testing

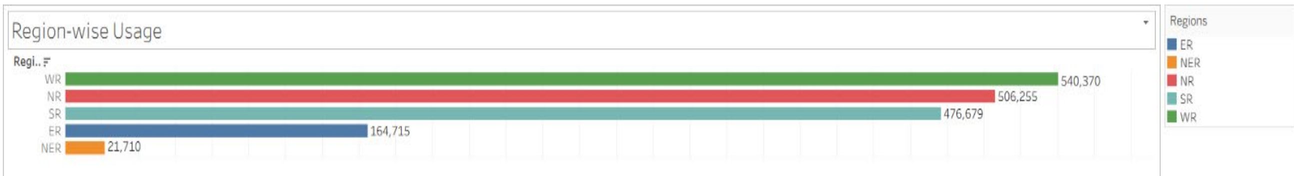
Successfully tested on:

- Windows
- Google Chrome
- Microsoft Edge
- MySQL Workbench
- Tableau Desktop
- Tableau Public
- Flask

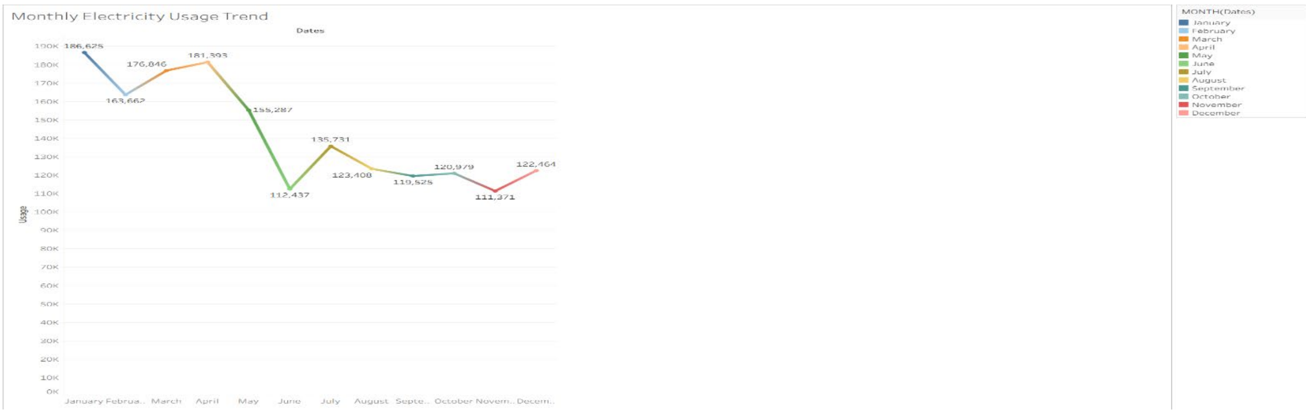
7. RESULTS



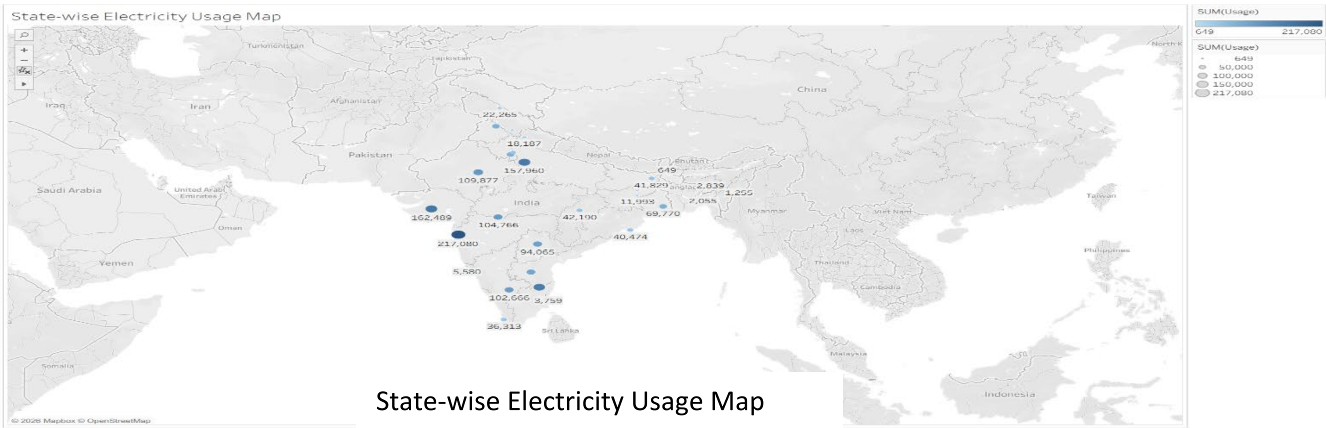
State wise Usage



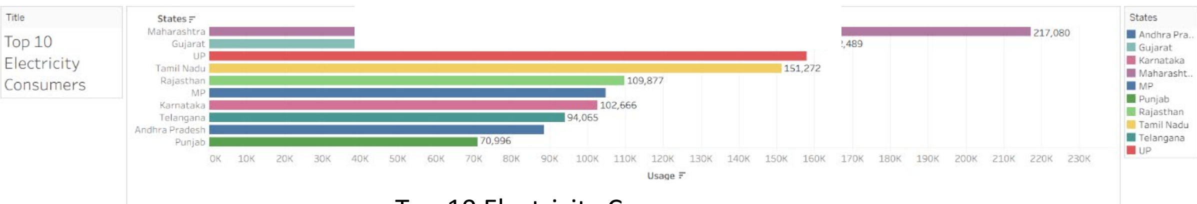
Region wise Usage



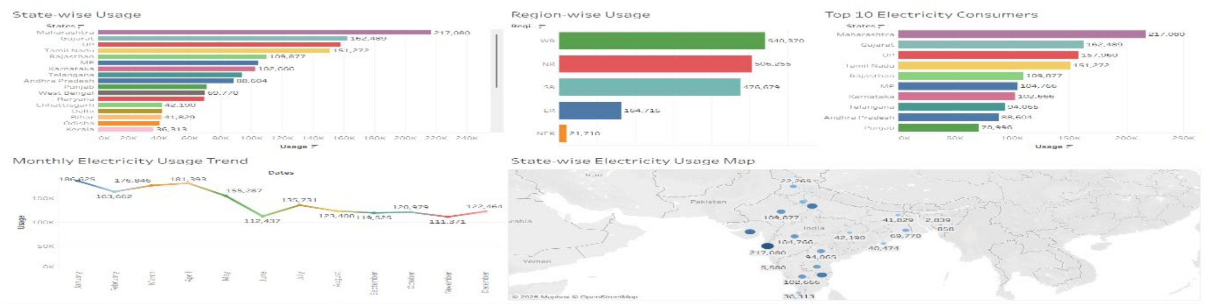
Monthly Electricity Usage



State-wise Electricity Usage Map



Top 10 Electricity Consumers



Dashboard

8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive dashboards.
- Faster analysis.
- Accurate SQL-based reporting.
- Easy comparison across states and regions.
- Web-based dashboard access.
- Supports data-driven decision-making.

Disadvantages

- Depends on the quality of the dataset.
- Requires periodic updates when new data becomes available.
- Performance may reduce with extremely large datasets.
- Requires internet access to view the Tableau Public dashboard.

9. CONCLUSION

The Electricity Consumption Analysis project successfully demonstrates how MySQL, Tableau, and Flask can be integrated to analyze and visualize electricity consumption data. The project provides interactive dashboards, geographical maps, monthly trend analysis, and business insights that support effective decision-making and efficient electricity management.

10. FUTURE SCOPE

- Integration with real-time electricity monitoring systems.
- AI-based electricity demand forecasting.
- Predictive analytics using machine learning.
- IoT-enabled smart energy monitoring.
- Mobile application integration.
- Multi-year electricity consumption comparison.

11. APPENDIX

- **Dataset: -**
  https://drive.google.com/file/d/1JxIkHNwXxjFztKq7ad0_KtkukCqTckNy/view
- **GitHub Repository: -**
  https://github.com/Alisha40477/Electricity-Consumption-Analysis
- **Tableau Public**: -
  https://public.tableau.com/app/profile/mohammed.mahaboob.alisha.shaik/viz/Electricity_Consumption_Analysis_17837707990200/Dashboard1?publish=yes
- **Flask Web Application (Local): -**
  http://127.0.0.1:5000
- **SQL Database Name**: - electricity

12. Tools & Technologies Used:

- MySQL Workbench
- Tableau Desktop
- Tableau Public
- Python 3.12.7
- Flask
- Visual Studio Code
- Git & GitHub